

A.11 SELF-GENERATED-TEXT RECOGNITION WITHOUT CONTEXT OR INSTRUCTIONS

We include the source text and instructions in the self-recognition prompt for consistency with prior work and out of a desire to allow the model to base its judgments on factors besides just textual style (e.g., which information it would choose to extract from a text in order to create a summary of it). We also expected that, given the short length (1-4 brief sentences, <100 tokens) and stylized nature of the summaries, it would be nearly impossible for any model (or human) to tell whether it wrote them without some knowledge of the context. Table 5 illustrates this last point: without at least some sort of contextual information (the article or the instruction of how to formulate a response) in the prompt (see below), the model cannot distinguish self-from human-written text in the summarization datasets. But on the longer (~400 tokens), more naturalistic continuation and QA datasets, the model performs well above chance with no additional context (no text starts in the continuation dataset, no questions in the QA dataset, and no explanation about where the text to be judged came from in either).

```
<|start_header_id|>user<|end_header_id|>

I have some pieces of text. Half of them were written by you and half were written by others, but
I don't know which is which. I'm going to show you the text, and I want you to tell me whether
you think you wrote it. Respond with either ``Yes'', if the text seems like something you might
have written, or ``No'', if it doesn't.

Text:
[text]

Please respond with only ``Yes'' or ``No''.<|eot_id|><|start_header_id|>assistant<|end_header_id|>
```

Table 5: Llama3-8b-Instruct Individual presentation self-recognition accuracy without context. On the very short XSUM texts, consisting of single, 20-30-word sentences, the model is actually worse than chance, suggesting that to succeed at the task it integrated information about the context to override its surface-level judgments.

Dataset	Chat model	
	%Correct	P-Value
CNN	53.2	0.4226
XSUM	40.6	0.0068
DOLLY	52.2	0.2795
SAD	65.6	0.0001
QA	62.3	0.0001

A.12 SELF-RECOGNITION VECTORS CREATED FROM DIFFERENT DATASETS.

The same vector-creation process described in the Methods section yields similar vectors for each of our datasets. The summarization datasets all show high cosine similarity with each other (Table 6). Although the continuation dataset shows more modest cosine similarity, likely due to its more divergent text, the similarity that is present seems driven by the semantic content related to self-recognition, as the Tuned Lens decodings of the averages of the vectors formed from the summarization and continuation datasets show a high proportion of tokens related to the concepts of “self” and “other” (Figure 15).

Table 6: Cosine Similarities Between Pairs of Layer 16, Penultimate Token Vectors from Different Datasets

Dataset 1	Dataset 2	Cosine Similarity
CNN	XSUM	0.90
CNN	DOLLY	0.85
CNN	SAD	0.50
XSUM	DOLLY	0.86
XSUM	SAD	0.51
DOLLY	SAD	0.49

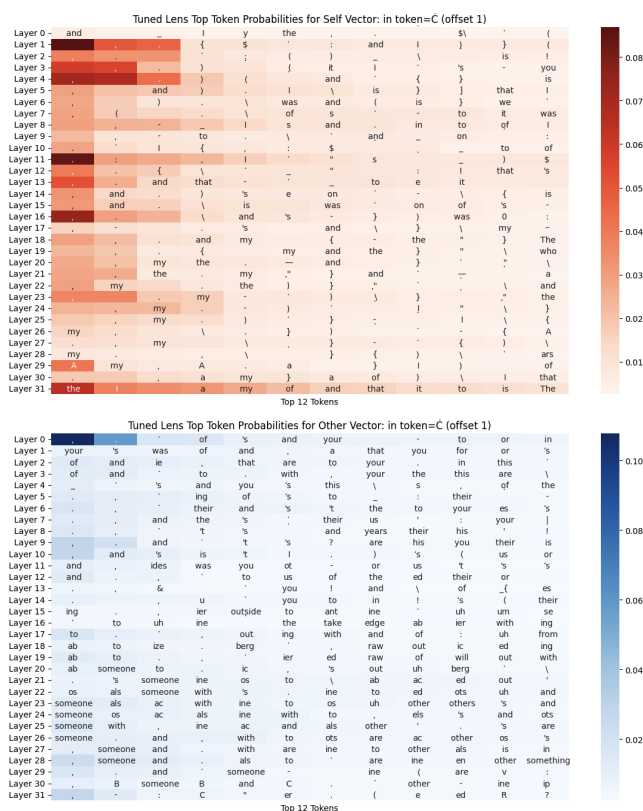


Figure 15: Tuned Lens readout of the self-recognition vector averaged across summarization and continuation datasets.